Artificial Intelligence Quiz Solutions

Q1: Apply the Hill Climbing algorithm starting from node "a" to reach node "h", selecting the child node with the lowest heuristic value at each step.

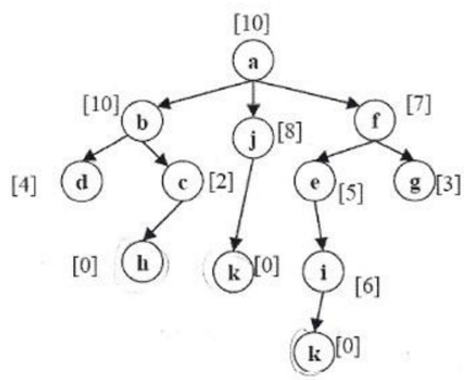

Figure 1: Graph for Hill Climbing Algorithm

Solution:

The Hill Climbing algorithm follows these steps:

1. Start at node **a** (heuristic = 10).
2. Choose the child with the lowest heuristic value:
   - b (10), j (8), f (7) → **Select f (7)**
3. From node **f**, choose the lowest heuristic child:
   - e (5), g (3) → **Select g (3)**
4. **g** has no child with a lower heuristic value than 3, so the algorithm is stuck at a local minimum.

Q2: A chess-playing AI competes against a human, analyzing the board state and making the best possible move based on future possibilities. Identify the environment type and agent type for this scenario.

Solution:

Environment Type:

- Competitive, deterministic, fully observable, discrete, and turn-based.

Agent Type:

- The chess AI is a **rational or goal-based,**.